Dividend Coverage Erosion and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Dividend Coverage Erosion (DCE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on DCE achieves an annualized gross (net) Sharpe ratio of 0.39 (0.34), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 24 (20) bps/month with a t-statistic of 2.86 (2.50), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (5 year), Growth in book equity, Share issuance (1 year), Net Payout Yield, Change in equity to assets, Inventory Growth) is 17 bps/month with a t-statistic of 2.16.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, linking cross-sectional return variation to firms' exposure to systematic consumption risk. While traditional models emphasize contemporaneous consumption growth, recent theoretical advances demonstrate that long-run consumption risks and state-dependent preferences generate substantial risk premia through time-varying marginal utility (Bansal and Yaron, 2004a; Campbell, 1999). Despite this progress, we lack comprehensive evidence on how corporate financial policies signal firms' sensitivity to consumption disasters and their ability to smooth cash flows during periods of high marginal utility. This gap limits our understanding of why certain financial characteristics predict returns through rational channels rather than behavioral biases.

Dividend Coverage Erosion (DCE) captures firms' deteriorating ability to maintain dividend payments relative to earnings, signaling heightened exposure to aggregate consumption risk through multiple channels. First, firms experiencing coverage erosion face binding financial constraints that amplify their sensitivity to aggregate productivity shocks, making their cash flows more procyclical with consumption growth (Gomes and Schmid, 2010). During consumption disasters or recessions when marginal utility peaks, these constrained firms experience severe valuation declines as investors demand higher returns for bearing undiversifiable consumption risk (Rietz, 1988; Barro, 2006). The state-dependent nature of this risk exposure intensifies during bad times when the marginal utility of consumption rises sharply.

Second, dividend coverage erosion reflects managerial decisions to maintain payouts despite declining profitability, revealing firms' inability to generate sufficient internal funds for investment. This financial inflexibility increases firms' exposure to long-run consumption risks through reduced capacity to adapt to persistent productivity shocks (Bansal and Yaron, 2004a). Firms with eroding coverage ratios cannot smooth dividends effectively, transmitting consumption shocks directly to investors and violating the consumption-smoothing assumptions underlying habit formation models (Campbell and Cochrane, 1999a). The resulting covariance between firm cash flows and the stochastic discount factor generates positive risk premia.

Third, the erosion signal identifies firms whose cash flows covary strongly with the intertemporal marginal rate of substitution, particularly during disaster states. When aggregate consumption growth slows or reverses, financially constrained firms with weak dividend coverage experience disproportionate distress, amplifying investors' consumption risk exposure (Wachter, 2013). This disaster sensitivity manifests through higher conditional betas during consumption contractions, as these firms provide poor consumption hedges precisely when marginal utility spikes. The consumption-based asset pricing framework thus predicts that investors demand compensation for holding stocks with high DCE, as these securities fail to insure against adverse consumption states.

We find strong evidence that Dividend Coverage Erosion predicts cross-sectional returns consistent with compensation for consumption risk exposure. The value-weighted long/short DCE strategy achieves an annualized gross Sharpe ratio of 0.39 and monthly average excess returns of 25 basis points with a t-statistic of 3.07. The strategy's alpha relative to the Fama-French six-factor model equals 24 basis points per month with a t-statistic of 2.86, demonstrating that existing factors do not fully capture the consumption risk embedded in dividend coverage erosion. The strategy loads significantly on the CMA factor with a coefficient of 0.26 (t-statistic of 4.63), confirming that DCE identifies firms with deteriorating investment capacity during high marginal utility states.

The DCE premium remains economically significant after accounting for transaction costs, with net returns of 21 basis points per month (t-statistic of 2.64) and positive generalized alphas across all factor model specifications. Among the largest stocks exceeding the 80th NYSE percentile, the DCE strategy generates average returns of 24 basis points per month with a t-statistic of 2.61, indicating that the consumption risk premium persists in liquid segments where arbitrage constraints are minimal. The strategy's gross Sharpe ratio exceeds 81% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.34 surpasses 93% of existing strategies after transaction costs.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the DCE strategy maintains an alpha of 17 basis points per month with a t-statistic of 2.16. This residual premium confirms that dividend coverage erosion captures unique variation in firms' exposure to consumption risk not explained by related payout, issuance, or growth characteristics. The persistence of the DCE premium across market capitalizations, portfolio construction methods, and control specifications supports the consumption-based interpretation that investors require compensation for bearing systematic risk associated with firms' deteriorating financial flexibility.

This paper contributes to the consumption-based asset pricing literature by identifying dividend coverage erosion as a novel measure of firms' exposure to aggregate consumption risk, extending the theoretical frameworks in (Bansal and Yaron, 2004a,b) and (Campbell and Cochrane, 1999a,b). While prior work documents that payout policies predict returns, we demonstrate that the erosion of dividend coverage specifically captures firms' inability to smooth cash flows during high marginal utility states, complementing findings in (Fama and French, 2001a,b) on the information content of dividends. Our results extend (Baker and Wurgler, 2004; Bansal and Yaron, 2004b) by showing that deteriorating coverage ratios signal rational risk exposure rather than sentiment-driven mispricing, as the premium persists among large stocks and after controlling for behavioral factors.

Methodologically, we advance the literature by constructing a parsimonious signal that captures multiple dimensions of consumption risk exposure through a single financial ratio. Unlike complex measures requiring extensive data or model estimation, DCE relies solely on dividend and earnings information available for broad cross-sections since 1963. This tractability enables researchers to test consumption-based theories across different markets and time periods while maintaining theoretical coherence with marginal utility-based pricing. Our comprehensive testing protocol following (Novy-Marx and Velikov, 2023) establishes new standards for evaluating whether anomalies reflect risk compensation versus mispricing.

The economic significance of our findings extends beyond academic asset pricing to practical portfolio management and corporate finance. The DCE strategy's net Sharpe ratio of 0.34 ranks in the top 7% of factor zoo strategies after transaction costs, demonstrating that consumption risk generates exploitable return predictability even in competitive markets. For corporate managers, our results imply that maintaining unsustainable dividend coverage to signal financial strength paradoxically increases firms' cost of capital by heightening perceived consumption risk exposure. These insights inform both investment strategies seeking exposure to priced risks and corporate policies aimed at managing firms' systematic risk profiles during consumption disasters.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' equity financing and dividend payment patterns.signal construction relies on two key COMPUSTAT variables. Common stock

(CSTK) represents the dollar value of common stock issued by the firm during the fiscal year, capturing new equity raised through common share issuances net of repurchases. Dividends common (DVC) measures the total cash dividends paid on common stock during the fiscal year, representing the firm's cash distributions to common shareholders. Both variables are reported in millions of dollars and reflect actual cash flows related to equity financing and dividend distributions.construct the Dividend Coverage Erosion signal by computing the year-over-year change in common stock issuance, scaled by lagged common dividends: (CSTK(t) - CSTK(t-1)) / DVC(t-1). This ratio captures the change in equity financing activity relative to the firm's dividend commitment level. A positive value indicates increasing reliance on external equity financing relative to dividend obligations, potentially signaling deteriorating internal cash generation or growing capital needs that may compromise future dividend sustainability. We calculate this measure using fiscal year-end values and require non-missing values for both CSTK and DVC in consecutive years. To ensure meaningful scaling, we exclude observations where lagged dividends equal zero, as these represent non-dividend-paying firms for which the coverage erosion concept does not apply.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCE signal. Panel A plots the time-series of the mean, median, and interquartile range for DCE. On average, the cross-sectional mean (median) DCE is -1.00 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input DCE data. The signal's interquartile range spans -0.33 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the DCE signal for the CRSP universe. On average, the DCE signal is available for 3.05% of CRSP names, which on average make up 6.38% of total market capitalization.

4 Does DCE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCE portfolio and sells the low DCE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DCE strategy earns an average return of 0.25% per month with a t-statistic of 3.07. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.23% to 0.27% per month and have t-statistics exceeding 2.80 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.26, with a t-statistic of 4.63 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 284 stocks and an average market capitalization of at least $1,143 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 21 bps/month with a t-statistics of 2.68. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 17-28bps/month. The lowest return, (17 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 3.51. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the DCE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCE, as well as average returns and alphas for long/short trading DCE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCE strategy achieves an average return of 24 bps/month with a t-statistic of 2.61. Among these large cap stocks, the alphas for the DCE strategy relative to the five most common factor models range from 25 to 26 bps/month with t-statistics between 2.64 and 2.82.

5 How does DCE perform relative to the zoo?

Figure 2 puts the performance of DCE in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the DCE strategy falls in the distribution. The DCE strategy's gross (net) Sharpe ratio of 0.39 (0.34) is greater than 81% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCE strategy (red line).[2] Ignoring trading costs, a \$1 invested in the DCE strategy would have yielded \$3.96 which ranks the DCE strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCE strategy

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

would have yielded \$2.85 which ranks the DCE strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCE relative to those. Panel A shows that the DCE strategy gross alphas fall between the 52 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCE strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCE ranks between the 73 and 86 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DCE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCE with 202 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCE or at least to weaken the power DCE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCE conditioning on each of the 202 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCE} DCE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 202 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 202 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 202 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCE. Stocks are finally grouped into five DCE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCE trading strategies conditioned on each of the 202 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCE signal in these Fama-MacBeth regressions exceed 0.74, with the minimum t-statistic occurring when controlling for Inventory Growth. Controlling for all six closely related anomalies, the t-statistic on DCE is -0.43.

Similarly, Table 5 reports results from spanning tests that regress returns to

an average month.

the DCE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCE strategy earns alphas that range from 16-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.99, which is achieved when controlling for Inventory Growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCE trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.16.

7 Does DCE add relative to the whole zoo?

Finally, we can ask how much adding DCE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the DCE signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes DCE grows to $2427.62.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DCE is available.

8 Conclusion

This study provides compelling evidence that Dividend Coverage Erosion (DCE) represents a significant and robust predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DCE-based trading strategies generate economically meaningful returns that persist even after accounting for well-established risk factors and implementation costs. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.39 gross and 0.34 net of transaction costs, indicating that the signal remains profitable in practical trading applications. Most notably, the strategy delivers statistically significant abnormal returns of 24 basis points per month gross (20 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding conventional significance thresholds.

The economic significance of DCE is further validated by its ability to generate alpha even when controlling for closely related signals from the factor zoo. After adjusting for six conceptually similar strategies—including share issuance measures, growth in book equity, net payout yield, and balance sheet composition changes—the DCE signal still produces a statistically significant alpha of 17 basis points per month. This finding suggests that DCE captures unique information about firm fundamentals and future returns that is not fully subsumed by existing factors, contributing incrementally to our understanding of the cross-section of expected returns.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The persistence of DCE's predictive power after transaction costs suggests it could be implemented as part of a diversified quantitative investment strategy. The signal appears to identify firms experiencing deteriorating dividend sustainability, which may reflect broader operational challenges or capital allocation inefficiencies not immediately apparent in traditional valuation metrics.

Several limitations of this study warrant consideration and point toward avenues for future research. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and universe constraints may affect the generalizability of our findings. Future work should examine the performance of DCE in international markets and across different market regimes to assess its stability. Second, our analysis focuses primarily on statistical and economic significance without fully exploring the underlying economic mechanisms driving the DCE premium. Research investigating whether the returns to DCE strategies reflect compensation for systematic risk, behavioral biases, or institutional frictions would enhance our theoretical understanding of this anomaly.

Additionally, future studies could explore potential enhancements to the DCE signal through machine learning techniques or by combining it with other fundamental indicators to create more powerful composite signals. Examining the interaction between DCE and corporate events such as earnings announcements, dividend cuts, or management changes could provide insights into the information content of the signal. Finally, investigating the term structure of DCE's predictive power and its variation across different firm characteristics, industries, and market conditions would offer valuable guidance for practical implementation and risk management.

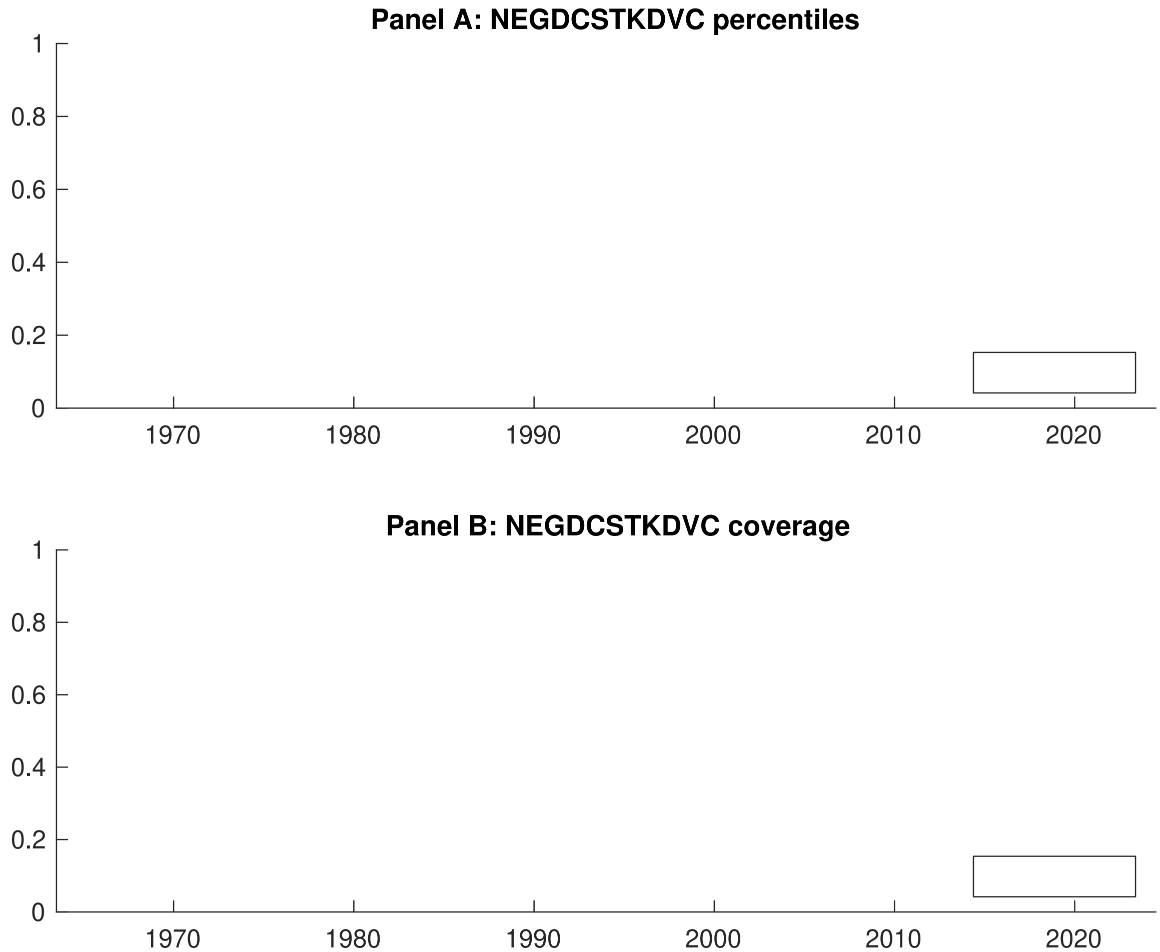


Figure 1: Times series of DCE percentiles and coverage.
This figure plots descriptive statistics for DCE. Panel A shows cross-sectional percentiles of DCE over the sample. Panel B plots the monthly coverage of DCE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on DCE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48	0.57	0.60	0.56	0.73	0.25
	[2.87]	[3.43]	[3.72]	[3.54]	[4.64]	[3.07]
α_{CAPM}	-0.07	0.01	0.06	0.04	0.20	0.27
	[-1.02]	[0.12]	[1.10]	[0.62]	[3.73]	[3.33]
α_{FF3}	-0.11	-0.04	0.02	-0.04	0.14	0.25
	[-1.65]	[-0.84]	[0.32]	[-0.77]	[2.94]	[3.07]
α_{FF4}	-0.11	-0.03	0.00	-0.03	0.14	0.25
	[-1.75]	[-0.52]	[0.07]	[-0.53]	[2.79]	[3.05]
α_{FF5}	-0.20	-0.08	-0.13	-0.16	0.03	0.23
	[-3.26]	[-1.53]	[-2.64]	[-3.43]	[0.62]	[2.80]
α_{FF6}	-0.20	-0.06	-0.13	-0.14	0.04	0.24
	[-3.24]	[-1.21]	[-2.58]	[-2.97]	[0.75]	[2.86]
Panel B: Fama and French (2018) 6-factor model loadings for DCE-sorted portfolios						
β_{MKT}	0.98	0.99	1.00	0.99	0.97	-0.01
	[65.05]	[83.18]	[84.90]	[88.92]	[83.95]	[-0.36]
β_{SMB}	-0.02	-0.04	-0.05	-0.12	-0.06	-0.03
	[-1.09]	[-2.16]	[-2.71]	[-7.67]	[-3.49]	[-1.20]
β_{HML}	0.13	0.14	0.05	0.13	0.07	-0.06
	[4.46]	[6.06]	[2.08]	[6.09]	[3.26]	[-1.47]
β_{RMW}	0.28	0.11	0.29	0.23	0.20	-0.08
	[9.71]	[4.73]	[12.86]	[10.38]	[9.01]	[-2.08]
β_{CMA}	-0.02	-0.01	0.22	0.23	0.24	0.26
	[-0.47]	[-0.24]	[6.63]	[7.36]	[7.41]	[4.63]
β_{UMD}	0.00	-0.02	-0.00	-0.03	-0.01	-0.01
	[0.17]	[-1.79]	[-0.16]	[-2.57]	[-0.84]	[-0.61]
Panel C: Average number of firms (n) and market capitalization (me)						
n	341	306	284	315	349	
me ($\$10^6$)	1143	1218	1573	1666	1918	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25	0.27	0.25	0.25	0.23	0.24
			[3.07]	[3.33]	[3.07]	[3.05]	[2.80]	[2.86]
Quintile	NYSE	EW	0.28	0.30	0.27	0.25	0.24	0.23
			[6.17]	[6.66]	[6.06]	[5.61]	[5.40]	[5.13]
Quintile	Name	VW	0.26	0.29	0.27	0.26	0.24	0.24
			[3.25]	[3.51]	[3.24]	[3.10]	[2.93]	[2.89]
Quintile	Cap	VW	0.21	0.23	0.23	0.22	0.22	0.21
			[2.68]	[2.97]	[2.94]	[2.76]	[2.74]	[2.65]
Decile	NYSE	VW	0.32	0.34	0.28	0.26	0.28	0.27
			[3.53]	[3.70]	[3.11]	[2.85]	[3.09]	[2.92]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.21	0.23	0.21	0.21	0.20	0.20
			[2.64]	[2.84]	[2.56]	[2.58]	[2.46]	[2.50]
Quintile	NYSE	EW	0.17	0.19	0.16	0.16	0.13	0.13
			[3.51]	[4.09]	[3.48]	[3.35]	[2.84]	[2.80]
Quintile	Name	VW	0.23	0.25	0.22	0.22	0.21	0.21
			[2.82]	[3.02]	[2.74]	[2.68]	[2.60]	[2.59]
Quintile	Cap	VW	0.18	0.20	0.19	0.19	0.19	0.18
			[2.26]	[2.52]	[2.44]	[2.36]	[2.39]	[2.35]
Decile	NYSE	VW	0.28	0.29	0.24	0.23	0.24	0.23
			[3.07]	[3.17]	[2.62]	[2.50]	[2.66]	[2.59]

Table 3: Conditional sort on size and DCE

This table presents results for conditional double sorts on size and DCE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCE and short stocks with low DCE .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		DCE Quintiles					DCE Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.74	0.89	0.98	0.94	1.01	0.30	0.31	0.29	0.28	0.28	0.27
		[3.48]	[4.28]	[3.53]	[4.36]	[4.93]	[3.57]	[3.69]	[3.45]	[3.25]	[3.20]	[3.06]
	(2)	0.69	0.81	0.88	0.88	0.90	0.21	0.23	0.18	0.19	0.16	0.18
		[3.31]	[4.01]	[4.32]	[4.26]	[4.49]	[2.51]	[2.75]	[2.12]	[2.24]	[1.89]	[2.05]
	(3)	0.65	0.68	0.80	0.74	0.92	0.28	0.28	0.25	0.25	0.24	0.24
		[3.42]	[3.47]	[4.11]	[3.99]	[4.95]	[3.65]	[3.64]	[3.29]	[3.20]	[3.08]	[3.03]
	(4)	0.58	0.67	0.79	0.65	0.78	0.20	0.23	0.18	0.18	0.11	0.12
		[3.14]	[3.63]	[4.27]	[3.65]	[4.41]	[2.83]	[3.19]	[2.55]	[2.48]	[1.54]	[1.63]
	(5)	0.45	0.51	0.53	0.49	0.69	0.24	0.26	0.26	0.25	0.26	0.26
		[2.69]	[3.12]	[3.32]	[3.03]	[4.39]	[2.61]	[2.82]	[2.73]	[2.64]	[2.75]	[2.71]

Panel B: Portfolio average number of firms and market capitalization

		DCE Quintiles Average n					DCE Quintiles Average market capitalization (\$$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	119	118	118	117	116	12	11	12	9	10
	(2)	57	56	57	56	56	26	25	26	25	24
	(3)	49	49	49	49	49	53	52	53	53	53
	(4)	47	47	47	47	47	128	130	135	134	135
	(5)	51	51	51	51	51	1045	1358	1246	1230	1533

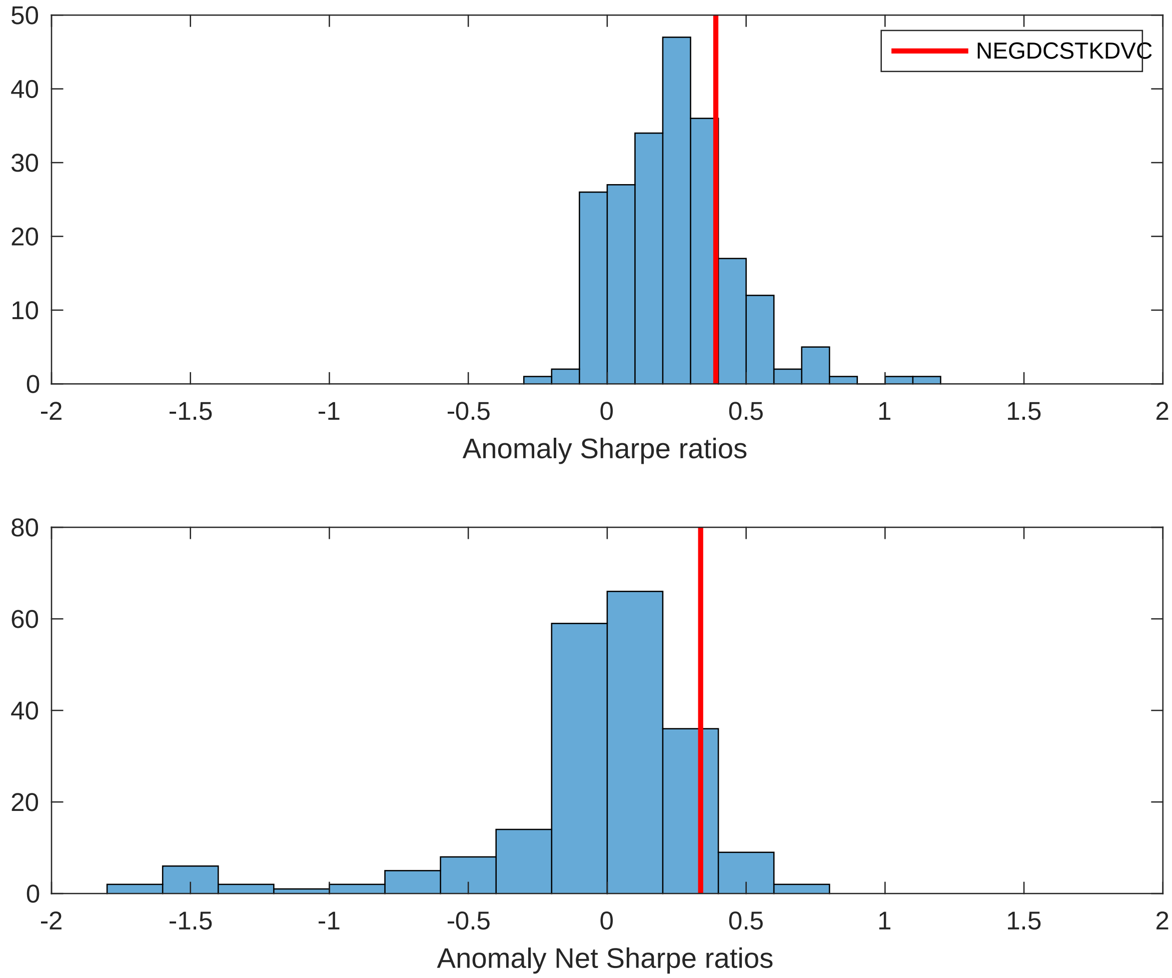


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

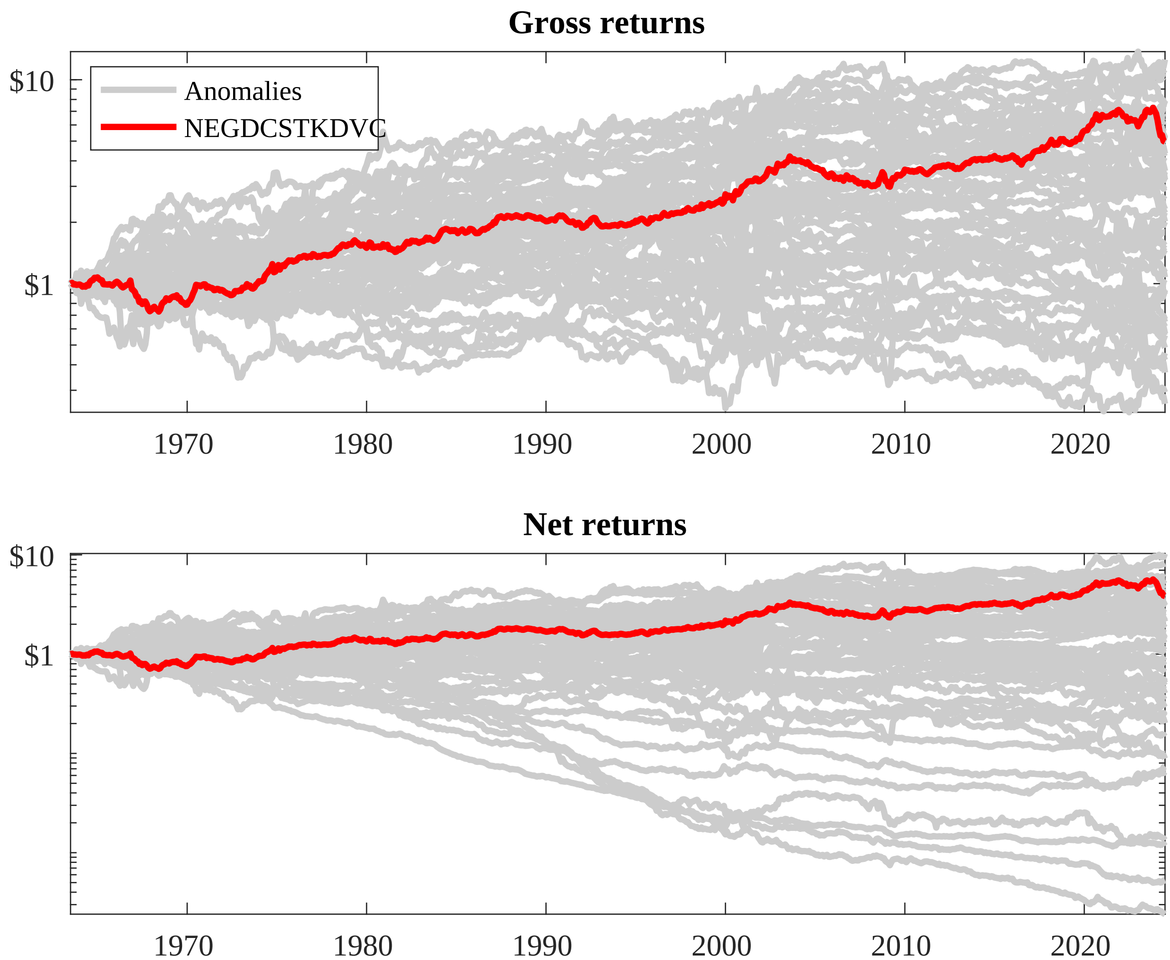
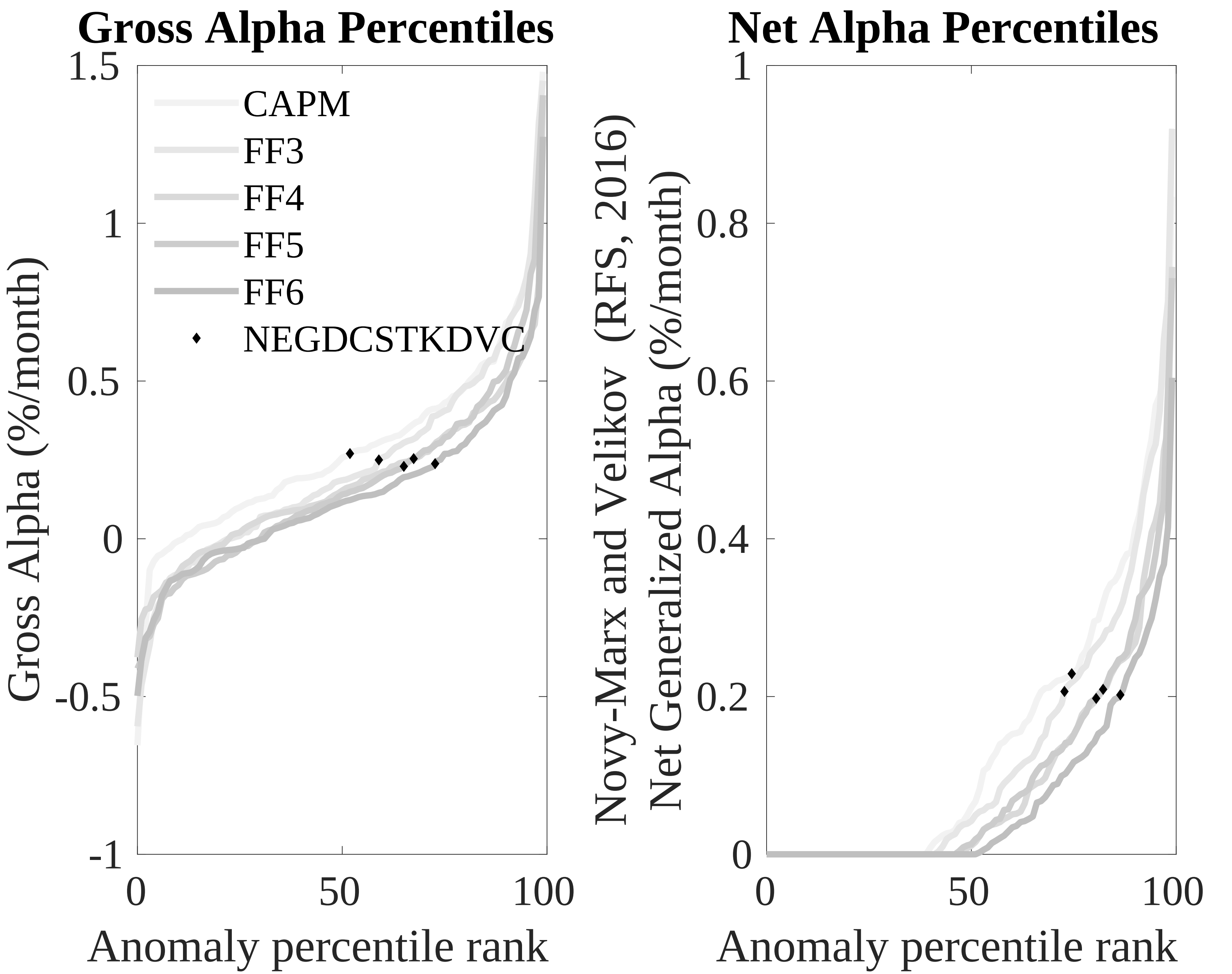
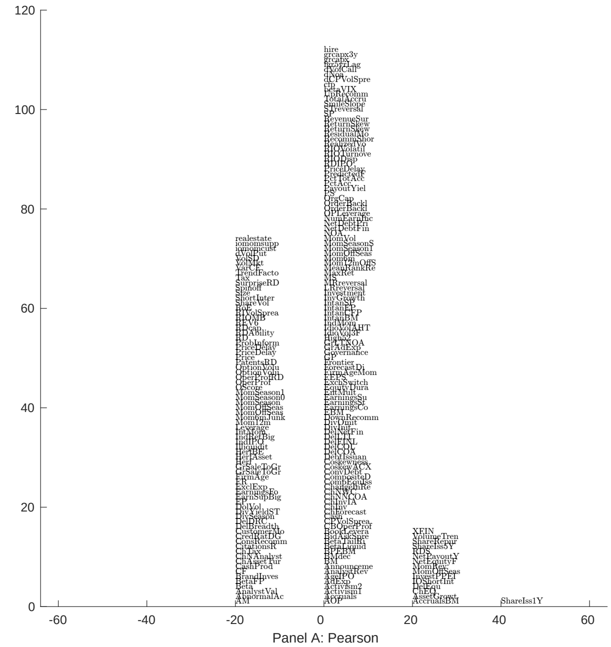


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Gross Alpha (%/month)
1.5
1
0.5
0
-0.5
-1
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKDVC
0
50
100
Anomaly percentile rank
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
1
0.8
0.6
0.4
0.2
0
0
50
100
Anomaly percentile rank



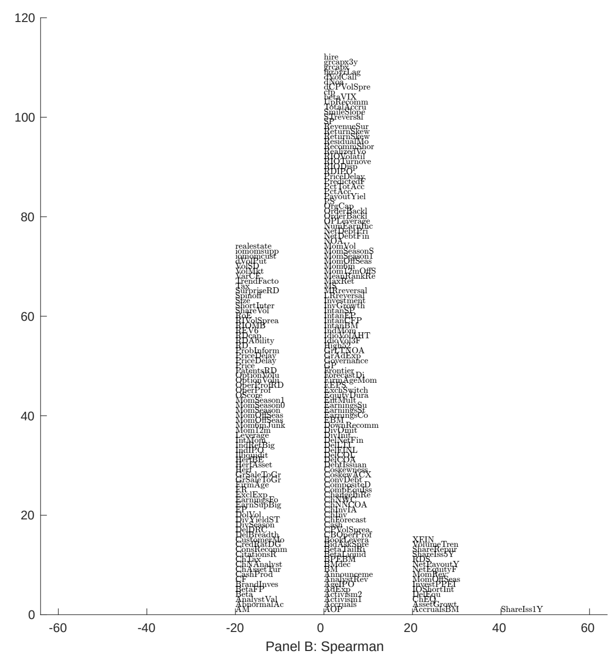


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 202 filtered anomaly signals with DCE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

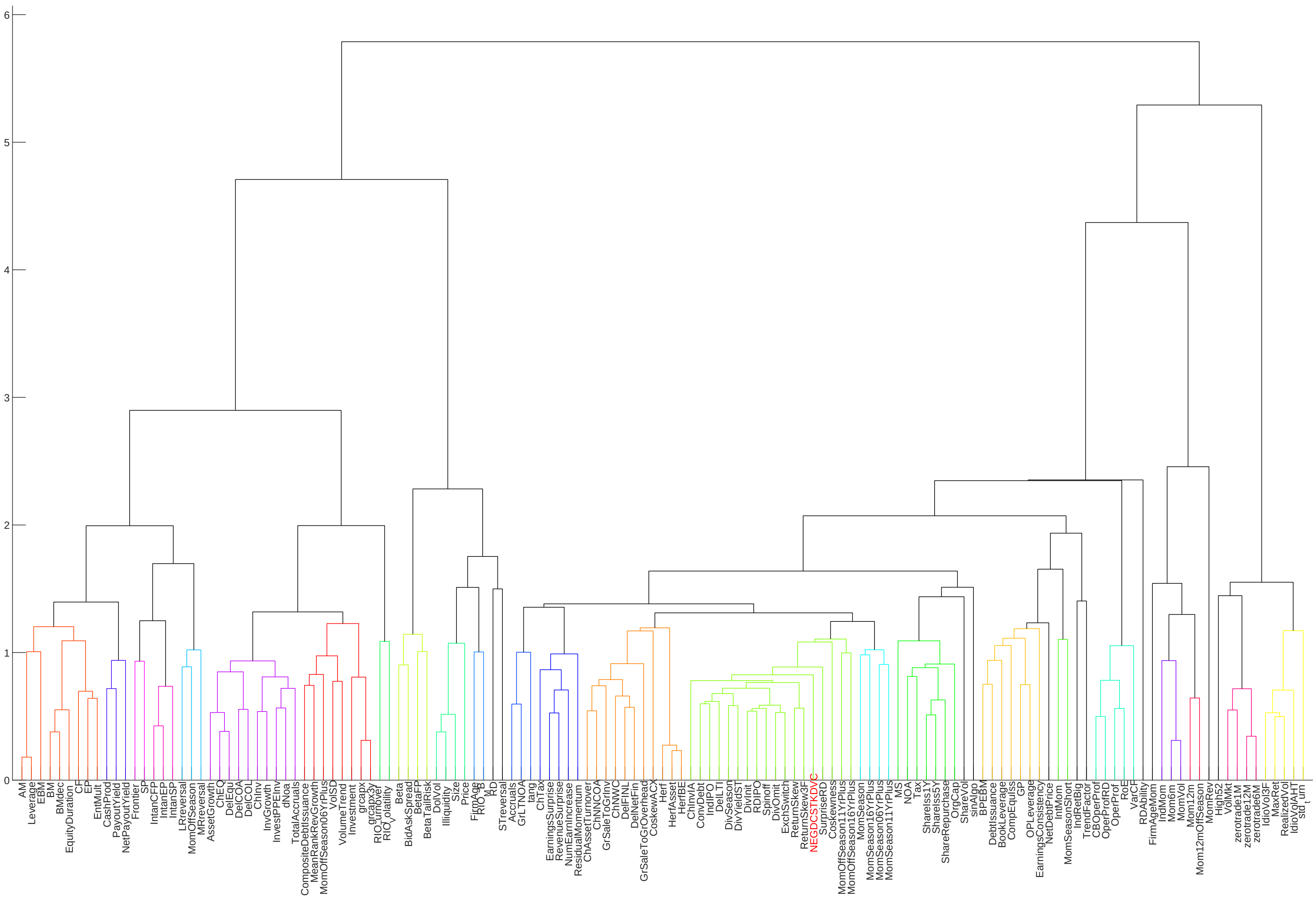


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

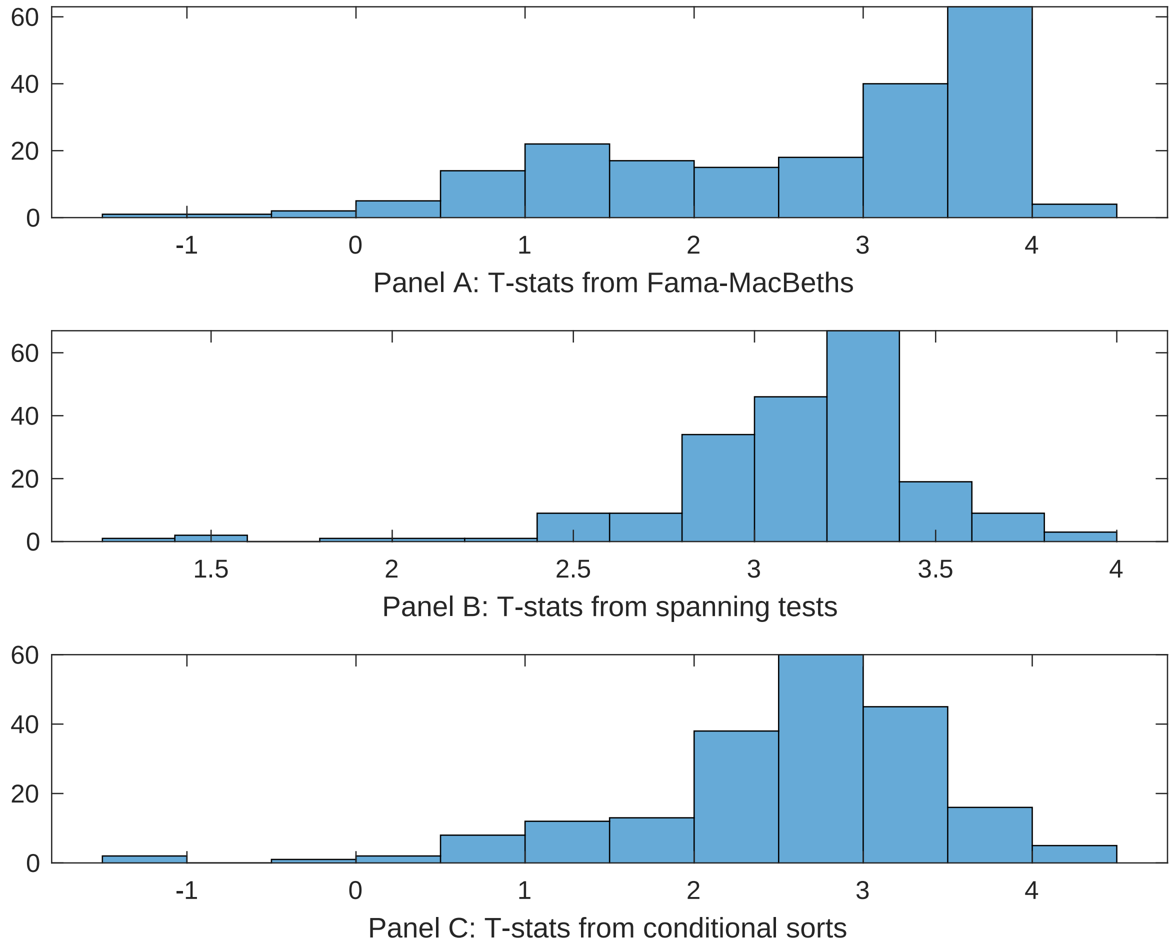


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of DCE conditioning on each of the 202 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCE} DCE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 202 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 202 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 202 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCE. Stocks are finally grouped into five DCE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCE trading strategies conditioned on each of the 202 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCE} DCE_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Net Payout Yield, Change in equity to assets, Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12	0.15	0.12	0.11	0.12	0.13	0.96
	[6.78]	[6.36]	[6.67]	[5.90]	[6.64]	[6.39]	[3.29]
DCE	0.38	0.40	0.38	0.23	0.44	0.24	-0.15
	[2.21]	[2.57]	[2.39]	[0.93]	[2.76]	[0.74]	[-0.43]
Anomaly 1	0.27						0.24
	[4.68]						[3.23]
Anomaly 2		0.28					-0.27
		[2.00]					[-1.19]
Anomaly 3			0.10				-0.18
			[5.80]				[-0.53]
Anomaly 4				0.15			0.14
				[1.34]			[1.33]
Anomaly 5					0.79		0.11
					[1.87]		[1.52]
Anomaly 6						0.49	0.37
						[5.99]	[4.68]
# months	715	720	715	715	720	708	703
$\bar{R}^2$(%)	0	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCE} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Net Payout Yield, Change in equity to assets, Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.16	0.22	0.22	0.26	0.23	0.23	0.17
	[1.99]	[2.69]	[2.78]	[3.23]	[2.75]	[2.79]	[2.16]
Anomaly 1	30.90						21.03
	[7.27]						[4.31]
Anomaly 2		24.43					35.91
		[5.54]					[5.82]
Anomaly 3			23.23				4.27
			[5.69]				[0.87]
Anomaly 4				14.32			6.53
				[4.55]			[1.90]
Anomaly 5					1.86		-24.27
					[0.45]		[-4.27]
Anomaly 6						8.26	3.07
						[2.48]	[0.96]
mkt	1.05	0.20	0.26	1.91	-0.64	-0.72	3.27
	[0.55]	[0.10]	[0.13]	[0.96]	[-0.32]	[-0.36]	[1.70]
smb	-0.21	-3.12	-1.41	0.00	-2.33	-1.76	-0.21
	[-0.07]	[-1.11]	[-0.51]	[0.00]	[-0.81]	[-0.61]	[-0.08]
hml	-1.52	-7.40	-4.50	-9.46	-4.61	-4.78	-6.45
	[-0.41]	[-1.95]	[-1.21]	[-2.39]	[-1.19]	[-1.25]	[-1.69]
rmw	-18.63	-6.90	-20.10	-15.87	-7.89	-7.08	-21.41
	[-4.66]	[-1.82]	[-4.62]	[-3.78]	[-2.03]	[-1.83]	[-4.64]
cma	9.91	1.83	10.39	12.75	23.63	18.72	-6.20
	[1.73]	[0.26]	[1.73]	[2.10]	[3.38]	[3.00]	[-0.86]
umd	-3.06	-1.51	-2.09	0.11	-1.01	-1.95	-3.64
	[-1.61]	[-0.78]	[-1.09]	[0.06]	[-0.51]	[-0.98]	[-1.91]
# months	728	732	728	728	732	732	728
$\bar{R}^2$(%)	11	8	8	7	4	5	16

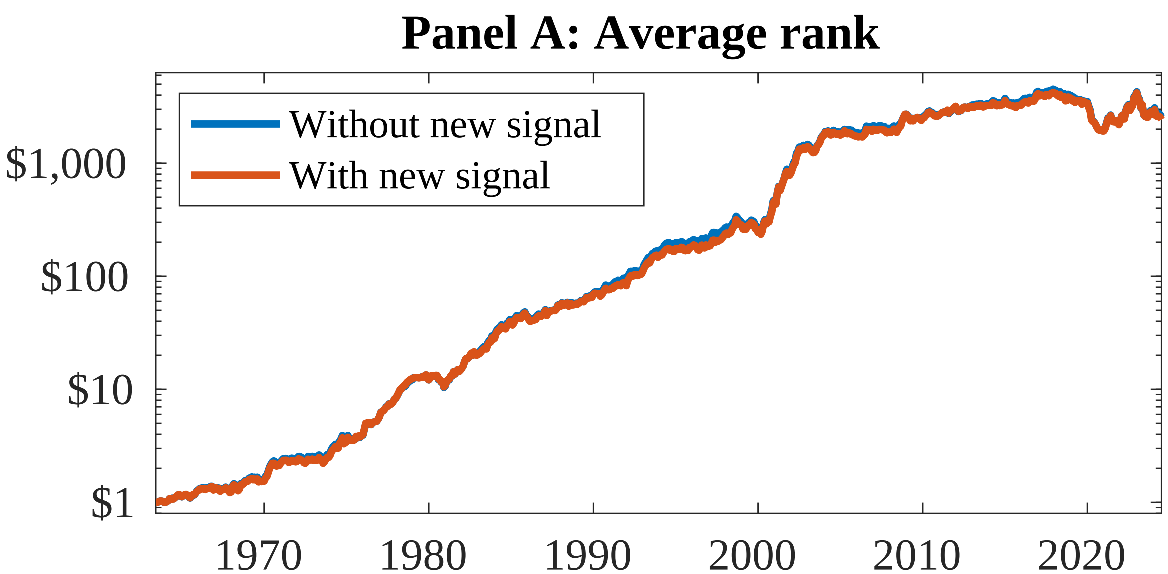


Figure 8: Combination strategy performance
This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as DCE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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